

VITALS

Score Report

Real model vs. baseline, per scenario

COLOR KEY

- real model
- constant_velocity baseline
- copy_last_state baseline

CONTENTS

occlusion_corridor – cosmos
occlusion_corridor – wan
ramp_descent_high_friction – cosmos
ramp_descent_high_friction – wan

cosmos

scenario: occlusion_corridor · n = 79 episodes · scored vs. baseline

cosmos
real model

constant_velocity
baseline

copy_last_state
baseline

VI₅₀ (median)

1.40s

CI [1.40, 1.40] · censor 0.00

Δ vs. constant_velocity

-0.93s

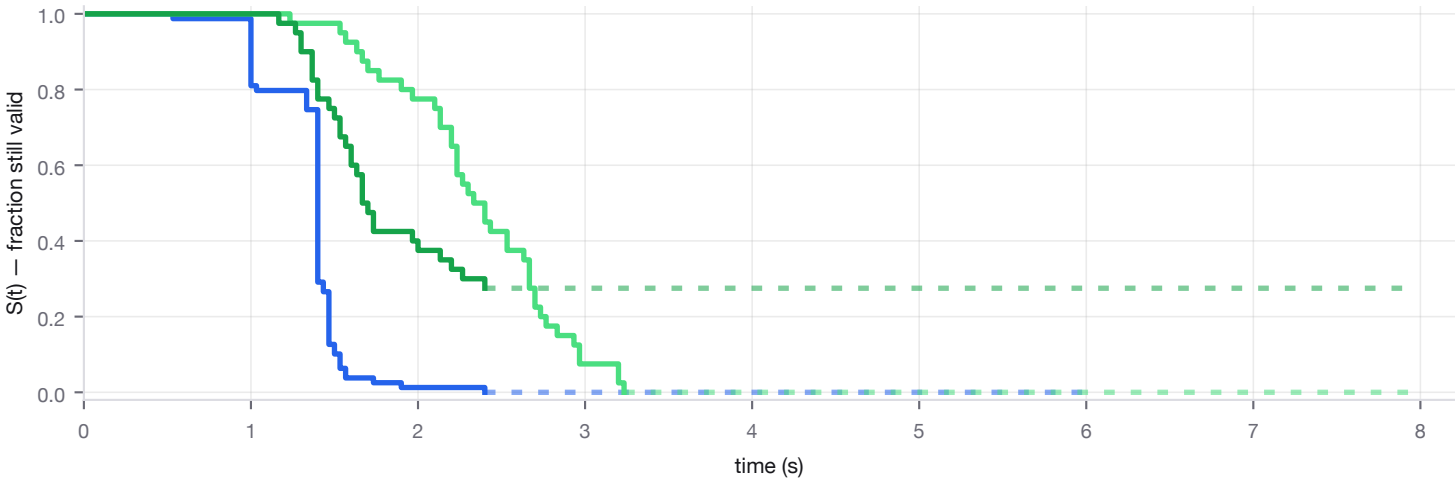
CI [-1.23, -0.80]s · worse (fails faster)

Δ vs. copy_last_state

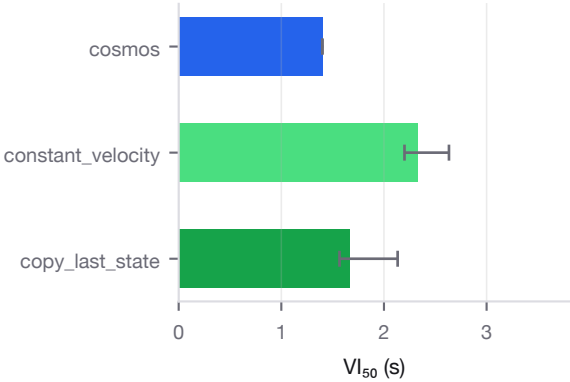
-0.27s

CI [-0.73, -0.17]s · worse (fails faster)

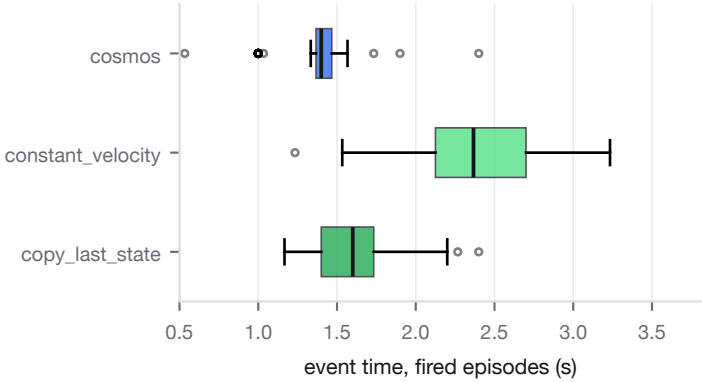
Degradation curve (Kaplan–Meier survival, dashed = censored past clip end)



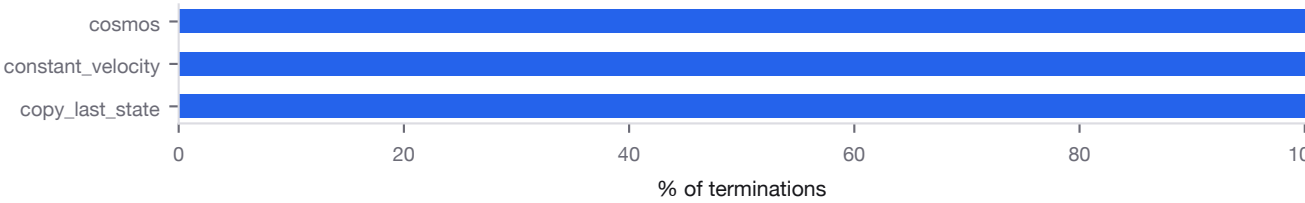
Median validity interval



Failure-time distribution



Failure mode (blue = R5 kinematic)



COMPUTED READOUT

```
beats_all_baselines = True
frac_failures_at_earliest_instant = 1.3%
dominant_failure_mode = R5 (100.0% of terminations)
vi50_delta[constant_velocity] = -0.93s, worse (fails faster)
vi50_delta[copy_last_state] = -0.27s, worse (fails faster)
seed_test_retest_median_abs_delta = 0.00s
minimum_detectable_difference(power>=80%) = 0.08s (VACUOUS -- zero/undefined natural spread, not a real threshold)
```

wan

scenario: occlusion_corridor · n = 5 episodes · scored vs. baseline

- wan

real model
- constant_velocity

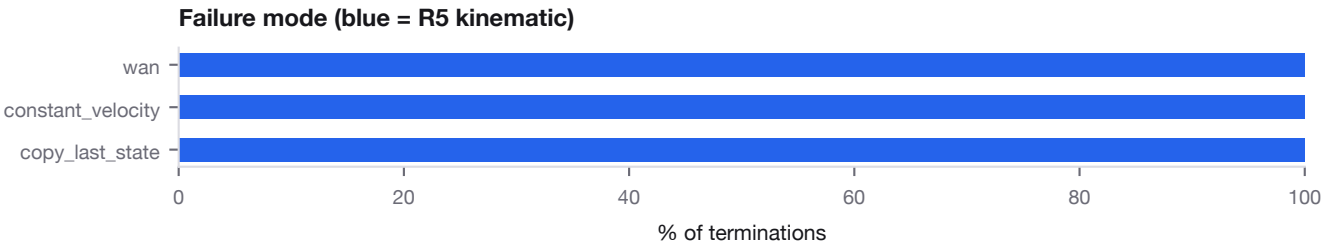
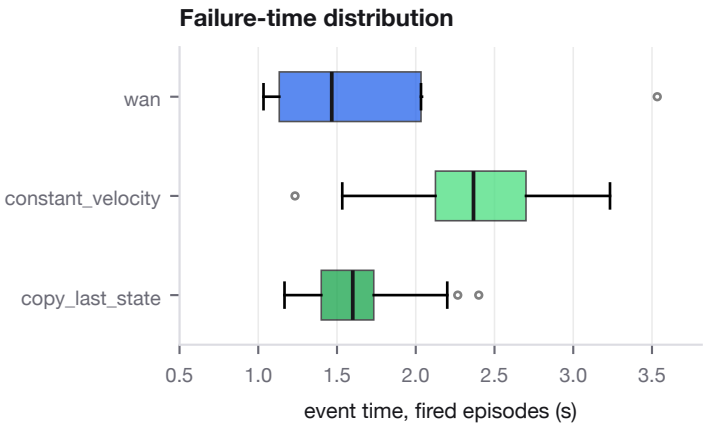
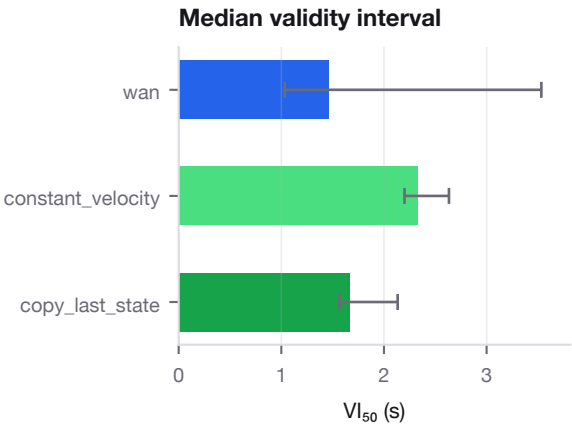
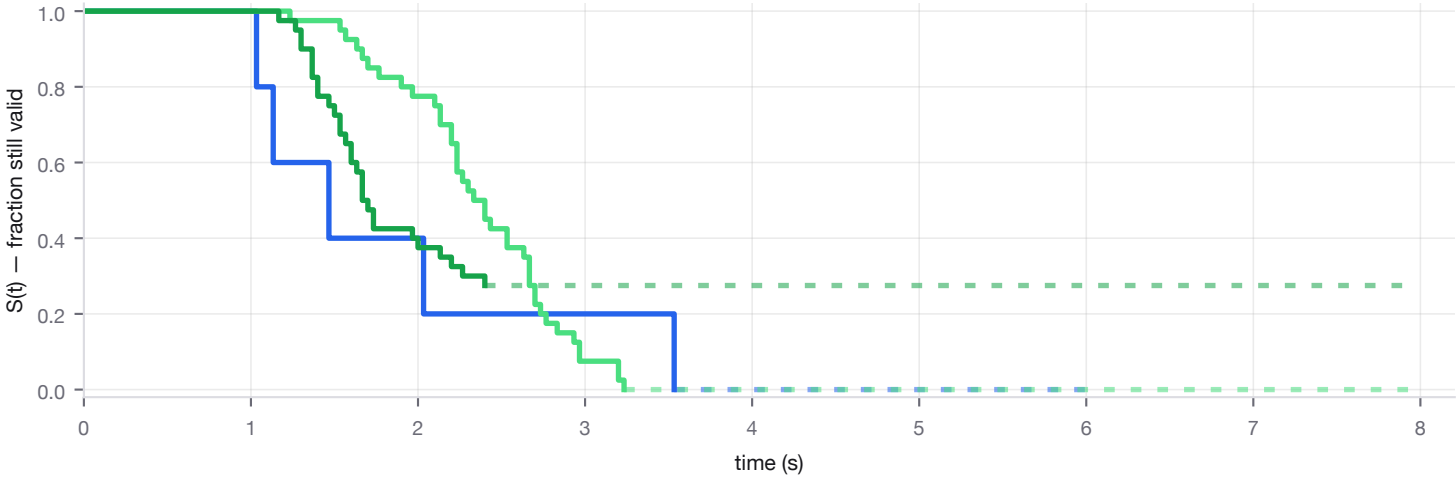
baseline
- copy_last_state

baseline

sample size n=5 is below the n=10 minimum for significance testing — figures below are illustrative only

VI₅₀ (median) 1.47s CI [1.03, 3.53] · censor 0.00	Δ vs. constant_velocity n/a insufficient sample	Δ vs. copy_last_state n/a insufficient sample
---	--	--

Degradation curve (Kaplan–Meier survival, dashed = censored past clip end)



COMPUTED READOUT

```
beats_all_baselines = True
frac_failures_at_earliest_instant = 20.0%
dominant_failure_mode = R5 (100.0% of terminations)
vi50_delta[constant_velocity] = insufficient sample
vi50_delta[copy_last_state] = insufficient sample
seed_test_retest_median_abs_delta = 0.33s
minimum_detectable_difference(power>=80%) = not reached within tested grid
```

cosmos

scenario: ramp_descent_high_friction · n = 80 episodes · scored vs. baseline

● cosmos

real model

● constant_velocity

baseline

● copy_last_state

baseline

VI₅₀ (median)

1.00s

CI [1.00, 1.00] · censor 0.00

Δ vs. constant_velocity

-1.40s

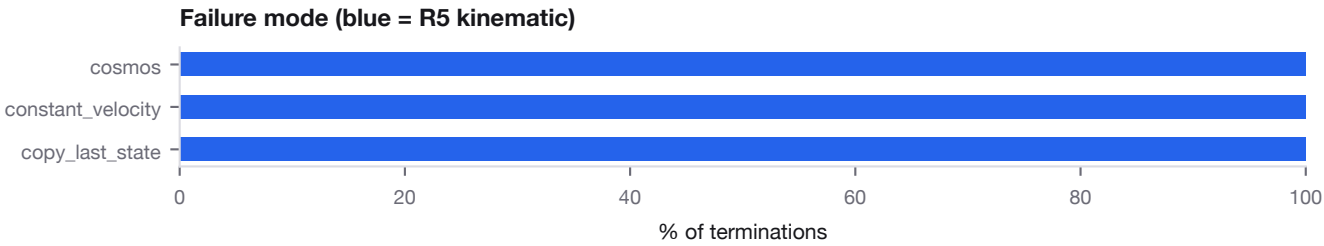
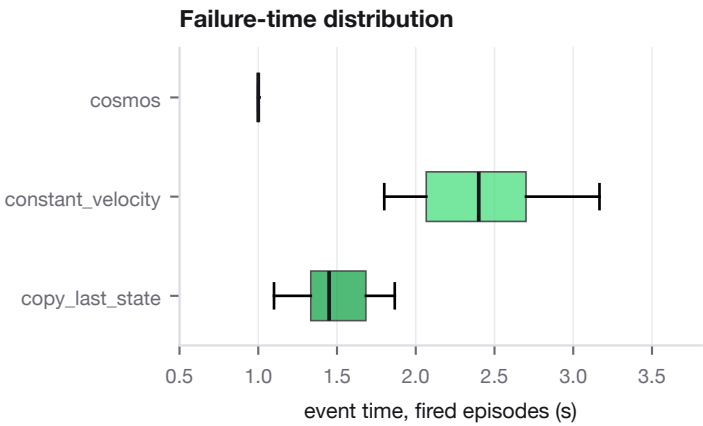
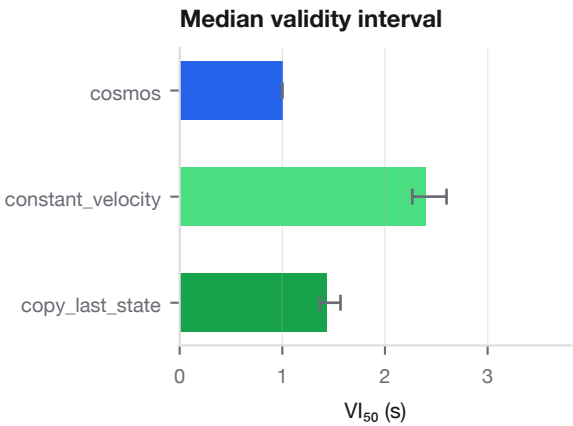
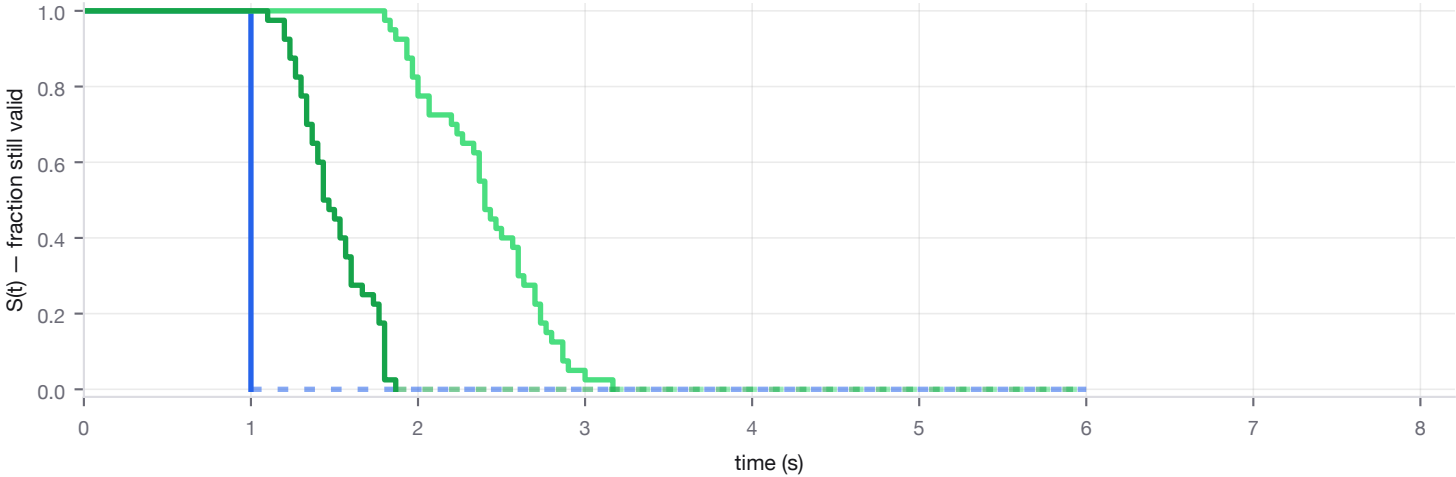
CI [-1.60, -1.27]s · worse (fails faster)

Δ vs. copy_last_state

-0.43s

CI [-0.57, -0.37]s · worse (fails faster)

Degradation curve (Kaplan–Meier survival, dashed = censored past clip end)



COMPUTED READOUT

```
beats_all_baselines = True
frac_failures_at_earliest_instant = 100.0%
dominant_failure_mode = R5 (100.0% of terminations)
vi50_delta[constant_velocity] = -1.40s, worse (fails faster)
vi50_delta[copy_last_state] = -0.43s, worse (fails faster)
seed_test_retest_median_abs_delta = 0.00s
minimum_detectable_difference(power>=80%) = 0.08s (VACUOUS -- zero/undefined natural spread, not a real threshold)
```

wan

scenario: ramp_descent_high_friction · n = 5 episodes · scored vs. baseline

- wan

real model
- constant_velocity

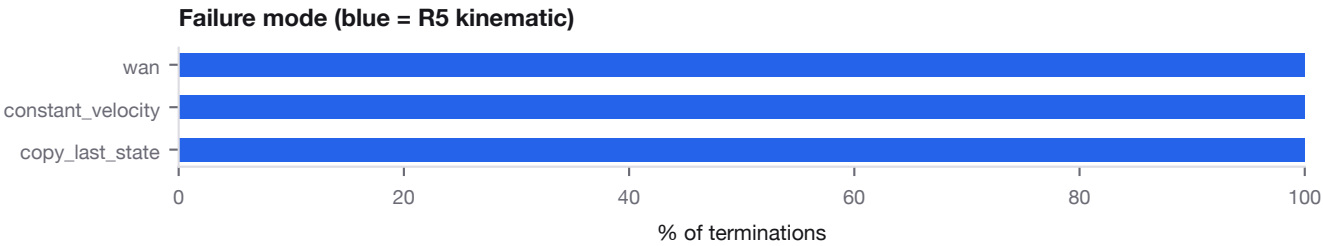
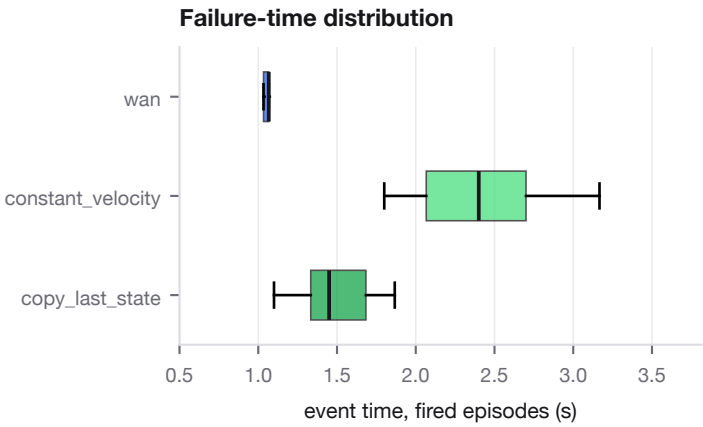
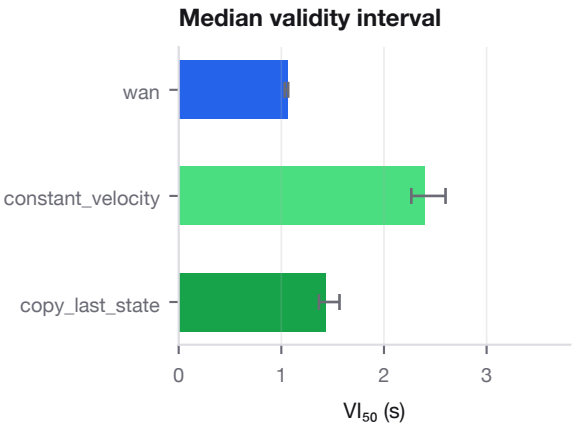
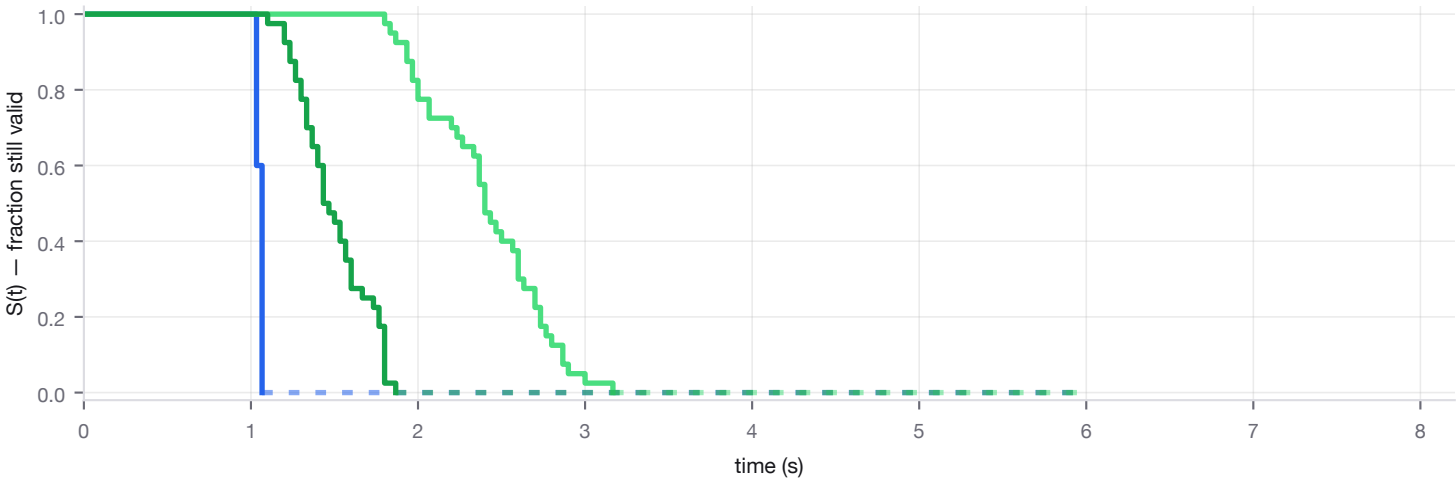
baseline
- copy_last_state

baseline

sample size n=5 is below the n=10 minimum for significance testing — figures below are illustrative only

VI₅₀ (median) 1.07s CI [1.03, 1.07] · censor 0.00	Δ vs. constant_velocity n/a insufficient sample	Δ vs. copy_last_state n/a insufficient sample
---	--	--

Degradation curve (Kaplan–Meier survival, dashed = censored past clip end)



COMPUTED READOUT

```
beats_all_baselines = True
frac_failures_at_earliest_instant = 40.0%
dominant_failure_mode = R5 (100.0% of terminations)
vi50_delta[constant_velocity] = insufficient sample
vi50_delta[copy_last_state] = insufficient sample
seed_test_retest_median_abs_delta = 0.03s
minimum_detectable_difference(power>=80%) = 0.05s
```